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SHORT-PAPER

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Published: 09 July 2018

Citation in BibTeX format

SETN '18: 10th Hellenic Conference on Artificial Intelligence
July 9 - 12, 2018
Patras, Greece

SETN '18: Proceedings of the 10th Hellenic Conference on Artificial Intelligence (July 2018)
https://doi.org/10.1145/3200947.3201041
ISBN: 9781450364331

Improving the Sequence Alignment Method by Quantum Multi-Pattern Recognition

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ABSTRACT

A dot matrix approach for sequence alignment is combined with a known quantum multi-pattern recognition method in order to improve the problem of sequence alignment. This dot matrix technique allows the application of some quantum computing principles on the pattern recognition problems like those used in the Grover's algorithm. When the recognized patterns exceed the limit of the 1/3 of the overall patterns, the multi-pattern recognition is accomplished simultaneously with the probability of 100%. The contribution concerns the way that the adopted quantum pattern recognition algorithm is applied speeding up the sequence alignment process. An example application demonstrates the effectiveness of the proposed method. String alignments isn't an essential task only in bioinformatics, but also in calculating the edit distance cost between strings in a natural language or in financial data.

CCS CONCEPTS

• **Theory of computation** → Pattern matching • **Computing methodologies** → Machine learning • **Applied computing** → Bioinformatics • **Computer systems organization** → Quantum computing

KEYWORDS

Quantum algorithm; Pattern recognition; Sequence Alignment; NP complete problem; Quantum parallelism

1 INTRODUCTION

Pattern recognition is a fast developing field, closely akin to machine learning, that focuses on the recognition of patterns or predictable recurrences within data, with interesting applications in areas such as bioinformatics. Multi-pattern recognition is an extension to incorporate more than one pattern at a time. Current quantum computing theory promises algorithms that can search for more than one pattern concurrently [1-2], so the concept of quantum multi-pattern recognition is going to be a legitimate field in quantum computing area.

This work focuses more on biological sequences, such as the alignment of genome sequences, referring to amino acid or nucleotide sequences, which is one of the fundamental operations for the study of life. A sequence may be synthesized by a specified set of elements. The term sequence alignment (SA) is defined as a way of arranging sequences in order to find similar regions between these sequences. Generally, SA is applied between two sequences (pairwise SA), but other methods may allow alignment for more than two sequences (multiple SA). Pairwise SA is accomplished based on dot matrix plots, dynamic programming, or heuristic methods. Today's dynamic programming methods may offer the optimal solution to the problem but in a NP way, especially for large scale alignments.

In this work, the proposed method is partially based on the quantum search algorithm introduced in [1]. A learning pattern set is initially defined to set the ground for the queries. Then, the adopted search algorithm is equipped with the training dataset and runs on the dot matrix datasets in order to identify one or more matching entries with 100% probability. The novelty is the way the dot matrix data are organized so as to allow the execution of the adopted search algorithm. This kind of preparation speeds-up the alignment process especially when a short sequence is compared to a large one (e.g. gene to chromosome sequences). Other approaches for single pattern SA treat the problem in a different way [3-4].

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SETN'18, July 2018, Rio Patras, Greece
© 2018 Association for Computing Machinery.
ACM ISBN 978-1-4503-6433-1/18/07...$15.00
https://doi.org/10.1145/3200947.3201041

The outline of this paper is organized as follows: In Section 2 the dot matrix SA method is described while in Section 3, some preliminary knowledge of quantum computation is shown. The proposed method is analyzed in Section 4 and Section 5 sums up this work.

2 DOT MATRIX SEQUENCE ALIGNMENT

In bioinformatics, there are mainly two ways of SA. The global, which covers the entire length of two roughly equal sequences (Needleman-Wunsch algorithm) [5] and the global which covers only parts of high similarity (Smith-Waterman algorithm) [6]. Both algorithms are based on dynamic programming and share the same concepts. The second interesting SA method is the Dot Matrix method.

This method is the most simple and qualitative one, though it is time-consuming in analyzing on a large scale. Certain sequence features -such as insertions, deletions, repeats, or inverted repeats of the elements of the sequence- can easily become identified by vision when a dot plot is formed for two sequences. The construction of such a plot demands the entry of the two sequences along the top row and the leftmost column of a two-dimensional matrix and the marking of the matching cross-points with dots. This matrix formation is similar to a typical recurrence plot. A real example is presented in Fig. 1 demonstrating the dot matrix plot. The dot plots of two similar sequences will appear as a single line along the matrix main diagonal. Moreover, when a protein consists of multiple similar structural domains the regions that share significant similarities will appear as lines off the main diagonal like those depicted in Fig. 2.

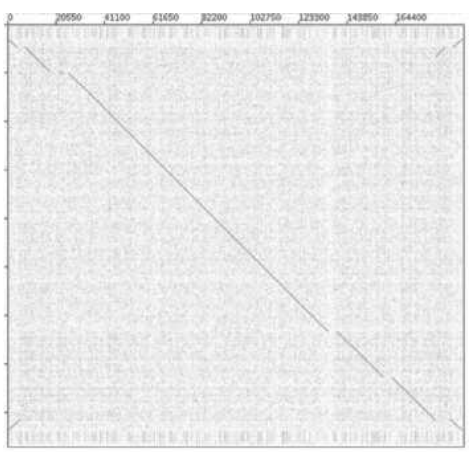

Figure 1: Dot plot of the two vaccinia's virus genomes: CVA and MVA. Large deletions are present in MVA.

3 QUANTUM COMPUTING BASICS

A quantum bit or qubit is a quantum system whose states lie in a two-dimensional Hilbert space. The state of a qubit is represented by Dirac bracket notation as $|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle$ since it may be in the ¨0¨, ¨1¨ or in both states simultaneously which is known as superposition. The complex numbers α and β determine the probability of the corresponding states and the unitarity is preserved $|a|^2 + |\beta|^2 = 1$. Closed quantum mechanical systems are transformed only via unitary transformations (quantum operators) that preserves quantum probabilities.

The state $|\varphi\rangle$ of a general quantum system can be expressed by the linear superposition of the basis states $|\varphi_i\rangle$ as the wave function $|\varphi\rangle = \sum_i c_i|\varphi_i\rangle$, where c_i the probabilities of each base state and $\sum_i |c_i|^2 = 1$. The basic notion in quantum computation is first to prepare the needed quantum states and then apply on them the proper quantum operators in order to bring the system to a final state where the solution can be extracted by measurement with high probability. The quantum measurement violates the coherence of the closed quantum system and the superposition is destroyed. The wave function $|\varphi\rangle$ governs the whole system and the result of a measurement will make the system to collapse to exactly one ground state $|\varphi_i\rangle$ with the corresponding probability c_i.

4 QUANTUM MULTI-PATTERN RECOGNITION METHOD

The proposed method neither rearranges sequences with the aid of a scoring matrix technique nor finds the optimal alignment solutions, such as in the aforementioned dynamical algorithms. Its purpose is to speed-up the pairwise SA process locating efficiently similar regions by using the adopted quantum pattern recognition algorithm proposed by R. Zhou and Q. Ding in [1]. Only a few runs of this algorithm are needed to return the indexes of all the similar regions for a particular set of patterns. When a lengthy sequence (e.g. chromosome) is aligned with a shorter one (e.g. gene), the performance may be proved more effective than the classical dot matrix technique. Thus, this method seems to be more appropriate for large scale SAs.

4.1 Data Formulation and Learning Pattern Set

Dot matrix method doesn't give us any detailed sequence information, as the user of this technique examines the plot visually, and only if the noise is absent. It is important first to find an efficient way to obtain the recurrence dot matrix, maintaining the ability to exploit diagonally the matrix. A sophisticated method in terms of a spatial light modulator (SLM) may allow us to obtain the recurrent plot more efficiently than the classical way [7].

The adopted quantum algorithm is similar to Grover's algorithm [8] in many ways. Hence, the search for the desired

patterns is conducted on a series of indexes that each one is connected to a full diagonal line's dataset of the dot matrix. A black box is considered to evaluate the equivalence between sequences, returning 1 when a matching element appears. The black box applies the comparison function between the sequences simultaneously in a parallel way (quantum parallelism). As an example, the red framed area of the recurring plot in Fig. 2 displays all the full-length diagonals that appear between the indexes 0 and 3075 of the horizontal (XL23808) sequence and the overall vertical (XLRHODOP) sequence. In order to compare the pursuit patterns with the diagonal lines appearing within the recurring plot, it is necessary to concern a single full-diagonal of the dot matrix as a distinct one-dimension array or vector dataset having the same length with the datasets of the target patterns. For the remaining unframed areas (with shorter diagonal's length) some extra paddings are added in their dataset to complete the searching process. Hence, the plot in Fig. 2 has a total of 4750 diagonal datasets, each of 1625 elements, to be identified by our adopted quantum pattern recognition algorithm.

Improving the Sequence Alignment Method by Quantum Multi-Pattern Recognition

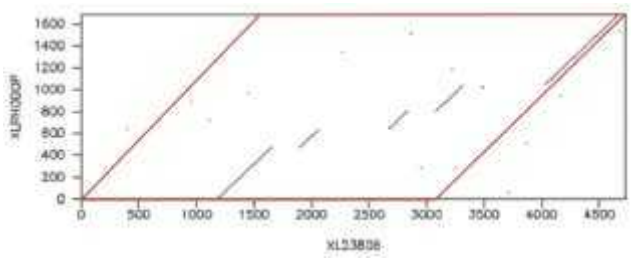

Figure 2: DNA sequence dot plot for the XL23808 and XLRHODOP sequences by the European Molecular Biology Laboratory

The learning pattern set should include training datasets of an equal size length to the smaller sequence of the pair under alignment. Achieving smaller diagonal lines results to smaller and fewer learning pattern sets. Hence, the method becomes more efficient when one of the sequences is small enough to be tolerated by the computing system.

In any case, a pattern and a spurious set are defined as $P = \{(x_1^n x_2^n \dots x_i^n)\}, x \in \{0,1\}$ and $S = \{(x_1^s x_2^s \dots x_i^s)\}, x \in \{0,1\}$, accordingly, where i is the number of the binary bit of a pattern and n or s is the total number of patterns in each set. The number of the patterns should satisfy the conditions $s = n$ and $s + n < 2^i$.

The formulation of the initial state $|\varphi_0\rangle$ of the system is given in Eq. 1 as derived by R. Zhou and Q. Ding [1]. The patterns in P and S are considered as the basis states of a linear superposition of states. The coefficients of the state $|\varphi_0\rangle$ that correspond to patterns of S and P are not zero, but the rest are regarded as equal to zero during the quantum encoding process.

$$|\varphi_0\rangle = \frac{1}{\sqrt{2n}} \sum_{x_i \in P,S}^{2n} |x_1 \dots x_i\rangle \tag{1}$$

This formulation accomplishes the learning of the pattern set.

4.2 Multi-Pattern Recognition Algorithm

The adopted recognition method is almost similar to the improved Grover search algorithm, as the only modification concerns the replacement of the target element by the trained initial $|\varphi_0\rangle$ state. The method is briefly described below:

(1) Form and encode $|\varphi_0\rangle$ into a linear superposition state by learning the pattern set.

(2) Apply the unitary operator to the quantum initializing state

$$U_{2n} = I - (1 - i) \sum_{i=1}^{2n} |x_i\rangle\langle x_i| \tag{2}$$

where I is the unity matrix and $|x_i\rangle$ is the index of the pattern and spurious states. This operator inverts the phases of $2n$ indices of states with 90°, but the phases of the other indices corresponding to the rest orthogonal states remain invariable.

(3) Apply the unitary operator to the quantum state

$$U_\varphi = (1 + i)|\varphi_0\rangle\langle\varphi_0| - \boldsymbol{i}I \tag{3}$$

where $\boldsymbol{i}$ is an imaginary number and $|\varphi_0\rangle$ is the index of the initial quantum state. This operator inverts the amplitudes of all indices about their average value.

(4) Steps 2 and 3 are executed iteratively $T \cong \frac{\pi}{4}\sqrt{N/n}$ times where $N=2^i$.

For large sequences, the initial encoding state demands that many qubits rendering hence, inflicting the implementation and the encoding process. A proposed solution is to divide the searching data into smaller fragments and apply the algorithm efficiently. Further classical algorithms can assemble the puzzle with the diagonals in the plot.

4.3 Example Application

In dot matrix plots the recognition patterns are the diagonal lines which may be imperfect. This method can recognize them with a probability of 100%, given that the learning pattern set is over 1/3 of the total patterns. Running the algorithm to locate the five diagonals in Fig. 2, presupposes that the parameter I (the length of the pattern set) isn't long enough in order to avoid complex encodings when preparing the initial state $|\varphi_0\rangle$. Thus, the vertical sequence is divided into smaller pieces of 50 digits each, which is a fordable number to accomplish initial encoding and it is long enough to locate acceptable similarities. This parameter can be optimized accordingly.

These settings lead to approximately 1600/50=32 separate applications of the searching method. The values of n and s can also be close to the 1/3 of the total patterns in order to maximize

the success probability. In this case, the number of the recognition patterns is quite large, equals to $2^{i=50}$=1,1259.10^{15} but the number of the recognition patterns is also very large since is the 1/3 of them. Thus, the algorithm is conducted $T \cong \frac{\pi}{4}\sqrt{N/n} = \frac{\pi}{4}\sqrt{1/3} \approx 1$ times per data section and the whole process requires about 32 running times. However, in each data section the algorithm may be executed more than once as more than one accepted diagonal lines may exist. The classical dynamic algorithms have a complexity of $O(s1.s2)$ in time, hence they need more than 7,2×10^6 computing steps. The parameters $s1$ and $s2$ are the two sequences.

5 CONCLUSIONS

The proposed method can spot the similar regions during a pairwise alignment in polynomial time and then it applies classical methods for local SA. This technique encourages the development of hybrid SA methods in order to avoid the construction of large substitution matrices or complex scoring methods of dynamic programming. The similarity degree can be easily parameterized by the user of the algorithm. The problem of data preparation by means of SLM is our future work.

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